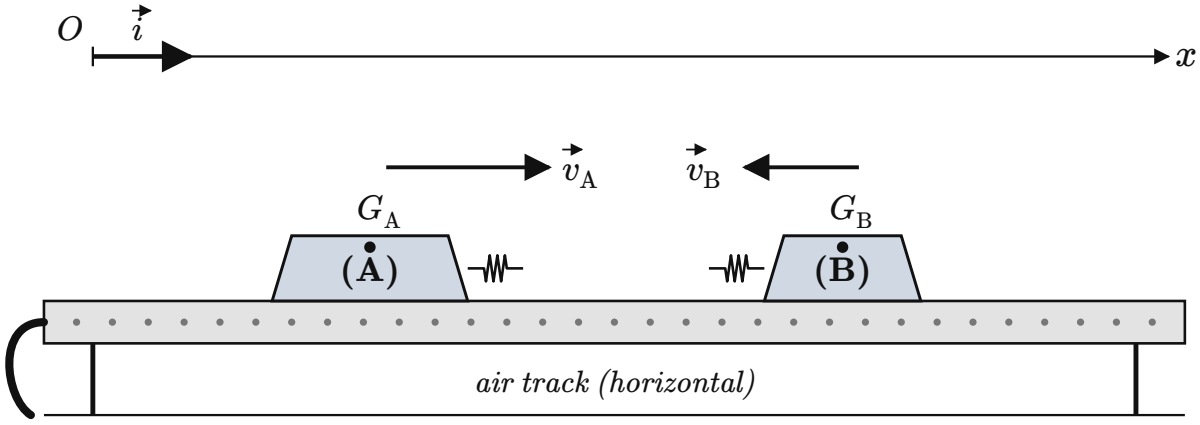
Grade 12 — Life Sciences Lebanese curriculum	**Practice Sheet** Chapter 2 — Linear Momentum	Duration: 2 hours Total: 20 points

This sheet is formed of three obligatory exercises. The use of a non-programmable calculator is allowed.

Take, in all exercises: $g = 10\ \text{m/s}^2$.

Exercise 1 — Testing bumpers on an air track (6 points)

In a laboratory, two gliders (A) and (B) are launched towards each other on a horizontal air track. Each glider carries, on the facing side, a light spring bumper, so that the two gliders bounce off each other without any permanent deformation (Doc. 1).



Doc. 1

Masses: $m_A = 300$ g; $m_B = 200$ g. Friction is negligible.
The centres of mass G_A and G_B of the gliders move along the axis $(O, \vec{i})$; x_A and x_B are their abscissae.
The bumpers touch at the instant $t = 0.40$ s. The duration of the contact is $\Delta t = 20$ ms; it is neglected in Doc. 2.
Doc. 2 gives the abscissae of G_A and G_B at several instants.

t (s)	0	0.20	0.40	0.60	0.80
x_A (cm)	0	10	20	18	16
x_B (cm)	40	35	30	43	56

Doc. 2

Part 1 — The collision

1.1 Show that the system [(A), (B)] is pseudo-isolated.

1.2 Using Doc. 2, show that each glider moves with a constant velocity before the collision and after it. Calculate the algebraic values v_A, v_B of the velocities before the collision and v'_A, v'_B after it.

1.3 Determine the linear momentum $\vec{P}$ of the system before the collision and $\vec{P}'$ after it. Conclude.

1.4 Show that the collision is elastic.

Part 2 — The centre of mass of the system

2.1 Write the expression of the abscissa x_G of the centre of mass G of the system in terms of m_A, m_B, x_A and x_B. Calculate x_G at the instants 0, 0.40 s and 0.80 s.

2.2 Deduce the nature of the motion of G and the value of its velocity v_G. Verify that $\vec{P} = (m_A + m_B)\,\vec{v}_G$.

Part 3 — During the contact

3.1 Calculate the variation of the linear momentum of each glider during the contact. Compare these two variations.

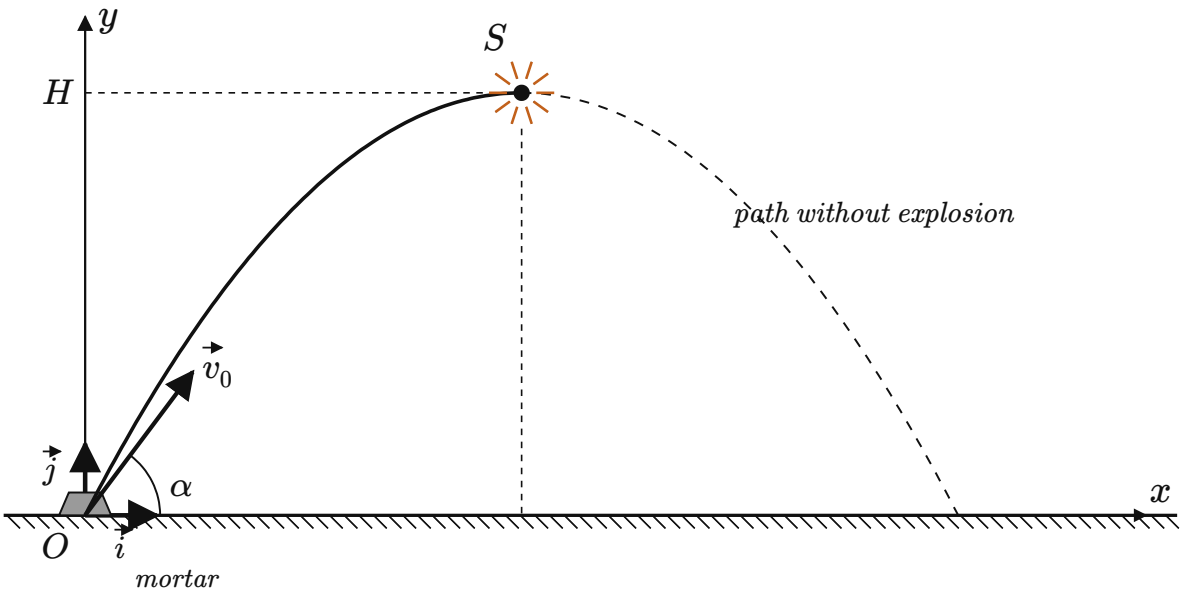
3.2 Taking $\dfrac{\Delta\vec{P}}{\Delta t}$ as equal to $\dfrac{d\vec{P}}{dt}$, determine the average force $\vec{F}_{B/A}$ exerted by (B) on (A) and the average force $\vec{F}_{A/B}$ exerted by (A) on (B). Which principle is verified?

3.3 At an instant of the contact, the bumpers are maximally compressed and the two gliders have the same velocity. Justify that this common velocity is $\vec{v}_G$, then determine the maximum elastic potential energy stored in the bumpers.

Exercise 2 — The fireworks shell **(7 points)**

During a celebration, a fireworks shell is fired from a mortar placed on the ground at O. It rises along a curved path and, at the highest point S, it explodes into two fragments (Doc. 3). Air resistance is neglected.

(not to scale)



Doc. 3

Mass of the shell: $M = 400$ g; the shell and its fragments are considered as particles.
At $t_0 = 0$, the shell leaves O with the velocity $\vec{v}_0$, of value $v_0 = 30$ m/s, making the angle α with the horizontal: $\sin\alpha = 0.8$; $\cos\alpha = 0.6$.
The motion is studied in the vertical plane $(O, \vec{i}, \vec{j}\,)$; the ground is the reference level for gravitational potential energy.

Part 1 — Before the explosion

1.1 Name the force acting on the shell after it leaves O. Is the shell an isolated system? Justify.

1.2 Applying Newton's second law, show that the horizontal component P_x of the linear momentum $\vec{P}$ of the shell is constant and that its vertical component is $P_y = Mv_0\sin\alpha - Mgt$. Calculate P_x and the value of P_y at $t_0 = 0$.

1.3 Deduce the expressions of the components v_x and v_y of the velocity of the shell in terms of time. Determine the instant t_1 at which the shell reaches S, and its linear momentum $\vec{P}_S$ at S.

1.4 Knowing that $v_y = \dfrac{dy}{dt}$ and that $y = 0$ at $t_0 = 0$, show that $y = 24t - 5t^2$ (SI). Deduce the height H of S and its abscissa x_S. Verify H using the conservation of the mechanical energy of the system [shell, Earth].

Part 2 — The explosion at S

At S, the shell explodes into two fragments: F_1 of mass $m_1 = 100$ g and F_2 of mass $m_2 = 300$ g. Just after the explosion, F_1 has a zero velocity.

2.1 The shell is not an isolated system. Explain why its linear momentum can nevertheless be considered as conserved during the explosion.

2.2 Determine the velocity $\vec{v}_2$ of F_2 just after the explosion (direction and value).

2.3 Calculate the kinetic energy of the system just before and just after the explosion. What is the origin of the difference?

Part 3 — Where do the fragments land?

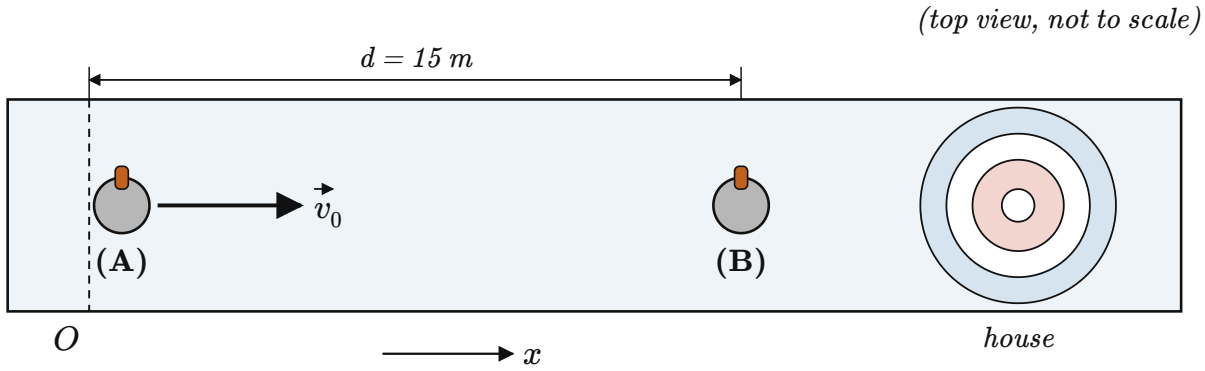
3.1 Show that the two fragments reach the ground at the same instant, 2.4 s after the explosion.

3.2 Determine the abscissae X_1 and X_2 of the points where F_1 and F_2 hit the ground.

3.3 Show that the centre of mass of the two fragments reaches the ground at the point where the shell would have landed without explosion. Interpret this result using Newton's second law applied to the system $[F_1, F_2]$.

Exercise 3 — A curling shot (7 points)

Curling is played on a horizontal sheet of ice. A player releases his stone (A) at O; it slides in a straight line and hits, head-on, a stone (B) of the other team, at rest at a distance $d = 15$ m from O (Doc. 4). The aim is to push (B) out and to keep (A) near the target, called the house.



Doc. 4

The two stones have the same mass $m = 20$ kg; they are considered as particles moving along the axis $(O, \vec{i})$.
While a stone slides, the ice exerts on it a friction force $\vec{f}$ of constant magnitude f, opposite to its velocity.
Doc. 5 gives the algebraic value P of the linear momentum of (A) from its release at $t_0 = 0$ until it hits (B) at $t_1 = 10$ s.



Doc. 5

Part 1 — The motion of (A) before the collision

1.1 Name the external forces acting on (A) during its sliding.

1.2 Using Doc. 5, write the expression of P in terms of time. Applying Newton's second law, determine the magnitude f of the friction force.

1.3 Deduce the expression of the algebraic value v of the velocity of (A) in terms of time and the value of its acceleration. Specify the nature of the motion of (A).

1.4 Calculate the speed v_1 of (A) just before the collision. Verify, by applying the kinetic energy theorem, that (A) travels the distance $d = 15$ m before hitting (B).

Part 2 — The collision

A video of the collision shows that the contact lasts $\Delta t = 2.0$ ms and that, just after it, (A) keeps moving in the same direction with the speed $v'_A = 0.10$ m/s.

2.1 The system [(A), (B)] is subjected to friction. Justify that its linear momentum may be considered as conserved during the collision.

2.2 Determine the velocity $\vec{v}'_B$ of (B) just after the collision.

2.3 Specify, with justification, the nature of the collision. What happens to the kinetic energy lost?

2.4 Determine the average force exerted by (A) on (B) during the contact. Deduce the average force exerted by (B) on (A).

2.5 Suppose now that the collision is elastic. Using the conservation of linear momentum and of kinetic energy, show that the two stones exchange their velocities.

Part 3 — Sweeping the ice

We go back to the real collision of Part 2. After it, (B) slides along $(O, \vec{i}\,)$; the centre of the house is 6.0 m beyond the point of collision.

3.1 Determine the distance travelled by (B) until it stops. Does (B) reach the centre of the house?

3.2 Players of the team of (B) sweep the ice in front of it, which reduces the friction force. Determine the magnitude of the friction force for which (B) stops exactly at the centre of the house.

— End of the sheet —